MIL-F-24202(SHIPS)
14 February 1966
SUPERSEDING
See 6.4

MILITARY SPECIFICATION

FITTINGS, BUTT WELDING, SEAMLESS

OR WELDED, 70-30 COPPER-NICKEL ALLOY,

700 PSI, 200°F.

1. SCOPE

1.1 Scope.- This specification covers 70-30 copper-nickel alloy seamless or welded butt-welding fittings in sizes 1/2 inch up to and including 16 inch iron pipe size (i.p.s.).

1.2 Types.- Fittings shall be of the following types, as specified (see 6.2):

Type I - Seamless
Type II - Welded

1.3 Pressure and temperature rating.- Fittings shall have a pressure rating of 700 pounds per square inch (p.s.i.) at 200°F.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-15726 - Copper-nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar and Plate) Forgings
MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded.

STANDARDS

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

FSC 4730

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STANDARDS

MILITARY (Cont'd)

MIL-STD-798 - Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler Construction Code.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street New York N.Y. 10017.)

AMERICAN STANDARDS ASSOCIATION (ASA)

B16.9 - Wrought Steel Butt Welding Fittings.

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, New York 10016.)

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification.- The fittings furnished under this specification shall be products which are qualified for listings on the applicable qualified products list at time set for opening of bids (see 4.3 and 6.3).

3.2 Materials and processing.

3.2.1 Fittings shall be made from pipe, tubing, forging, bar stock or plate by extruding, forming, forging, machining, fusion welding or by a combination of two or more of these operations.

3.2.2 Fittings machined from forged bars are acceptable. Fittings machined directly from rolled bars are not acceptable unless at least 1/3 of the diameter of the bar as rolled is removed from the center of the bar in the machining process.

3.2.3 Fittings made from pipe or tubing shall meet the chemical and physical requirements, with the exception of the hardness requirement, of 70-30 copper-nickel alloy in accordance with MIL-T-16420.

3.2.4 Fittings made from bars, plates or forgings shall meet the chemical and physical requirements of 70-30 copper-nickel alloy in accordance with MIL-C-15726, except as modified herein.

3.2.5 All finished fittings shall be furnished in the fully annealed condition. Annealing shall be performed at 1300° ± 50°F per hour per inch of thickness.

3.3 Design.-

3.3.1 General. - All fittings furnished under this specification shall conform to the requirements of ASA B16.9 and the additional requirements of 3.3.1.1 and 3.3.1.2.

3.3.1.1 Reinforcement.- All side outlets, such as tee branch connections, shall be adequately reinforced. Reinforcement areas shall be determined by any recognized design method.

3.3.1.2 Caps.- Caps shall be ellipsoidal in shape and shall conform to the requirements as given in ASME Boiler Construction Code.

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3.4 Dimensions.-

3.4.1 Unless otherwise specified herein, all dimensions are in inches.

3.4.2 Fitting dimensions shall be in accordance with figures 1 through 15 and tables II through VII and on approved drawings.

3.4.3 At no point throughout the fitting, shall the wall thickness be less than dimension "T" shown on figure 2 (see table II). For reducing fittings, the minimum wall thickness at any location shall be based on the external diameter at that location.

3.4.4 All elbows, and return bends shall have a tangent on each end in accordance with column D of table II.

3.4.4.1 When fittings are ordered for consumable insert welding (see 6.2), the tangent may be omitted. The center-to-end dimension for fittings intended for consumable insert welding shall be as shown in table II less the length of the tangent.

3.4.5 Tolerances on dimensions shall be in accordance with table III.

3.4.6 Dimensional and contour changes.- Any modifications to fitting design involving dimensional or contour changes other than weld end preparation, will require a retest in accordance with 4.3.2 prior to acceptance.

3.5 Fabrication.-

3.5.1 All welding shall be in accordance with section 7 of MIL-STD-798.

3.5.2 Fittings shall contain no more than one weld, either longitudinal or circumferential. The weld shall not be located in any of the principal axes as shown on figure 1.

3.5.3 Unless otherwise specified in the contract or order, all fittings shall be furnished with weld end details in accordance with figure 2.

3.6 Marking.-

3.6.1 Each fitting shall be legibly and permanently marked in accordance with section 10 of MIL-STD-798 and 3.6.2 and 3.6.3 herein.

3.6.2 Marking shall include manufacturer's name or trademark, size (IPS) 700 WOG, type I (for seamless) or type II (for welded), manufacturer's heat code identification number and MIL-F-24202.

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3.6.3 The required markings shall be 1/8 to 1/2 inch high depending on the size of the fitting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The fittings are considered to be "Level I" (see MIL-STD-798) items for the purpose of quality assurance and shall be manufactured under a quality control system which is in accordance with the requirements of section 8 of MIL-STD-798.

4.3 Qualification tests 1/ Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification approval may be obtained as specified in 4.3.1 or 4.3.2.

4.3.1 Fittings with wall thickness shown in tables VI and VII.- One sample size 4 inch tee fitting with the dimensions shown in tables II through V and the wall thickness shown in tables VI and VII shall be satisfactorily burst tested in accordance with the requirements of ASA B16.9. Manufacturers production drawings with a report of the burst test results for the size 4 inch tee fitting and the manufacturers production drawings for all other sizes for which qualification approval is desired shall be forwarded to the Bureau of Ships for approval and retention.

4.3.2 Fittings with wall thickness shown in table II.- One sample fitting of each type and each size and wall thickness shown in tables II through V shall be burst tested in accordance with the requirements of ASA B16.9 In the case of reducing fittings a satisfactory test of a straight fitting which is the same size as the largest connection of the reducing fitting shall eliminate the need of testing the reducing fittings. Failure of associated test equipment, such as connecting piping below the burst test pressure, shall require retesting. The following data shall be submitted to the Bureau of Ships for approval in connection with the burst tests:

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3 and 6.3.1).

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(a) Burst test results for each sample fitting, including a complete history of the material and strength characteristics of the fittings that were tested.

(b) Dimensional history of the sample design fitting including the design sketch of the unit.

(c) Manufacturers production drawings which show all pertinent dimensions taken from the sample fittings (as minimum) against which each fitting manufactured shall be checked by the manufacturer's inspection department.

4.4 Quality conformance inspection.

4.4.1 Lot.- All fittings of the same type and size offered for delivery at one time shall be considered a lot for the purpose of sampling.

4.4.2 Sampling for visual and dimensional examination.- A random sample of fittings shall be selected from each lot in accordance with table I for the examination specified in 4.4.3. If the number of defective fittings in any sample exceeds the acceptance number for that sample the lot represented by the sample shall not be offered for delivery.

Table I - Sampling for visual and dimensional examination

Number of fittings in lot	Number of fittings in sample	Nonconforming or defective fittings	
		Acceptance No.	Rejection No.
1 to 5	All	0	1
6 to 50	5	0	1
51 to 150	20	1	2
151 to 280	32	2	3
281 to 500	50	3	4
501 to 1200	80	5	6
1201 to 3200	125	7	8
3201 and up	200	10	11

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4.4.3 Visual and dimensional examination.- Each of the sample fittings selected in accordance with 4.4.2 shall be visually and dimensionally examined to verify compliance with all the requirements of this specification not involving tests and with the approved drawings. As a minimum, each sample fitting shall be examined for adequate thickness at the locations shown on figures 11 through 15 and delinated by "X". Each section so marked shall have the thickness checked at four locations around the circumference, 90 degrees apart except for the measurement of the crotches. For tee crotches, the measurement shall be taken as shown on figure 13. The numbers shown on figure 13 indicate the number and location of inspection points.

4.4.4 Nondestructive inspection.-

4.4.4.1 Dye penetrant inspection.- Dye penetrant inspection shall be performed in accordance with MIL-STD-278. Dye penetrant inspection shall be conducted as follows:

(a) Butt weld end preparation - All sizes
(b) External surfaces - All sizes
(c) Internal surfaces - 2 inches and above.

All defects in butt weld preparation found when dye penetrant inspected must be removed prior to acceptance. Only linear defects, as defined in MIL-STD-798, must be removed from other surfaces prior to acceptance. Repair welding shall be in accordance with section 7 of MIL-STD-798.

4.4.4.2 Radiographic inspection.- All welds shall be 100 percent radiographed in accordance with section 7 of MIL-STD 798. Acceptance criteria shall be in accordance with the requirements of section 7 of MIL-STD-798.

4.5 Material control - All manufacturers producing these fittings shall have a material and control system in accordance with section 9 of MIL-STD-798.

4.6 Material identification.- Material identification shall be in accordance with section 10 of MIL-STD-798.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.-

5.1.1 Preservation and packaging.- Preservation and packaging which may be the suppliers commercial practice shall be sufficient to affort adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation. Weld end preparation areas shall be protected from damage during shipment.

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5.1.1.1 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.2 Marking - ~~Shipment~~ marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal Stock Number or manufacturer's part number, contract or order number, contractor's name and destination.

5.2 Domestic shipment and storage or overseas shipment.- The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.- Fittings shall be preserved and packaged level A or C in accordance with MIL-V-3.

5.2.1.2 Packing.- Fittings shall be packed by level A, B or C in accordance with MIL-V-3.

5.2.1.3 Marking.- In addition to any special marking required, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use.- Fittings covered by this specification are intended primarily for shipboard water and gas service.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Type required (see 1.2)
(c) Whether tangents may be omitted (see 3.4.4.1)
(d) Preservation, packaging, packing and marking requirements, if other than 5.1 (see 5.2)

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(e) Size required (see tables II through VII)
(f) Drawings required
(g) Quantity required

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids qualified for inclusion in Qualified Products Lists 24202 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington D.C. 20360, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

6.4 Supersession data.- This specification replaces Bureau of Ships Drawing 810-1385879.

Preparing activity:
Navy - SH
(Project 4730-N092Sh)

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Table II - General dimensions (all dimensions are in inches).

Size	Pipe	T	F	D	A		B	C	E	P	K
IPS	O.D.	Min. 'C' bore thickness	Min. Counter-bore	Min. Tangent	90° elbows Long radius	90° elbows Short radius	45° elbows long radius	Tees straight	Caps	180° return	180° return
1/2	0.840	0.072	5/16	5/16	1-13/16	--	15/16	---	---	---	---
3/4	1.050	.083	5/16	5/16	1-7/16	--	3/4	1-1/8	---	---	---
1	1.315	.095	5/16	5/16	1-13/16	1-5/16	1-3/16	1-1/2	1-1/2	3	2-1/2
1-1/4	1.660	.095	5/16	5/16	2-3/16	1-9/16	1-5/16	1-7/8	1-1/2	3-3/4	3-1/16
1-1/2	1.900	.109	5/16	5/16	2-9/16	1-13/16	1-7/16	2-1/4	1-1/2	4-1/2	3-9/16
2	2.375	.120	5/16	5/16	3-5/16	2-5/16	1-11/16	2-1/2	1-1/2	6	4-1/2
2-1/2	2.875	.134	5/16	5/16	4-1/16	2-13/16	2-1/16	3	1-1/2	7-1/2	5-1/2
3	3.500	.165	5/16	5/16	4-13/16	3-5/16	2-5/16	3-3/8	2	9	6-9/16
3-1/2	4.000	.180	1/2	1/2	5-3/4	4	2-3/4	3-3/4	2-1/2	10-1/2	7-3/4
4	4.500	.203	1/2	1/2	6-1/2	4-1/2	3	4-1/8	2-1/2	12	8-3/4
5	5.563	.220	1/2	1/2	8	5-1/2	3-5/8	4-7/8	3	15	10-13/16
6	6.625	.259	1/2	1/2	9-1/2	6-1/2	4-1/4	5-5/8	3-1/2	18	12-13/16
8	8.625	.340	1/2	1/2	12-1/2	8-1/2	5-1/2	7	4	24	16-13/16
10	10.750	.380	1/2	1/2	15-1/2	10-1/2	6-3/4	8-1/2	5	30	20-7/8
12	12.750	.454	1/2	1/2	18-1/2	12-1/2	8	10	6	36	24-7/8
14	14.000	.473	1/2	1/2	21-1/2	14-1/2	9-1/4	11	6-1/2	42	28-1/2
16	16.000	.534	1/2	1/2	24-1/2	16-1/2	10-1/2	12	7	48	32-1/2

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Table III - Tolerances

IPS	A,,B, C, M and H ±	E ±	P ±	K ±	U ±
1/2 to 4-1/2	1/16	1/8	1/4	1/4	1/32
5 to 9	1/16	1/4	1/4	1/4	1/32
10 to 16	1/32	1/4	3/8	1/4	1/16

All dimensions are in inches.

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Table IV - Concentric and eccentric reducer dimensions
(all dimensions are in inches)

IPS	H
1 x 3/4	2
1 x 1/2	2
1-1/4 x 1	2
1-1/4 x 3/4	2
1-1/4 x 1/2	2
1-1/2 x 1-1/4	2-1/2
1-1/2 x 1	2-1/2
1-1/2 x 3/4	2-1/2
1-1/2 x 1/2	2-1/2
2 x 1-1/2	3
2 x 1-1/4	3
2 x 1	3
2 x 3/4	3
2-1/2 x 2	3-1/2
2-1/2 x 1-1/2	3-1/2
2-1/2 x 1-1/4	3-1/2
2-1/2 x 1	3-1/2
3 x 2-1/2	3-1/2
3 x 2	3-1/2
3 x 1-1/2	3-1/2
3 x 1-1/4	3-1/2
3-1/2 x 3	4
3-1/2 x 2-1/2	4
3-1/2 x 2	4
3-1/2 x 1-1/2	4
3-1/2 x 1-1/4	4
4 x 3-1/2	4
4 x 3	4
4 x 2-1/2	4
4 x 2	4
4 x 1-1/2	4
5 x 4	5
5 x 3-1/2	5
5 x 3	5
5 x 2-1/2	5
5 x 2	5
6 x 5	5-1/2
6 x 4	5-1/2
6 x 3-1/2	5-1/2
6 x 3	5-1/2

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Table IV - Concentric and eccentric reducer dimensions
(all dimensions are in inches)(Cont'd)

IPS	H
6 x 2-1/2	5-1/2
8 x 6	6
8 x 5	6
8 x 4	6
8 x 3-1/2	6
10 x 8	7
10 x 6	7
10 x 5	7
10 x 4	7
12 x 10	8
12 x 8	8
12 x 6	8
12 x 5	8
14 x 12	13
14 x 10	13
14 x 8	13
14 x 6	13
16 x 14	14
16 x 12	14
16 x 10	14
16 x 8	14

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Table V - Reducing tee dimensions (all dimensions are in inches)

IPS	M
1 x 1 x 3/4	1-1/2
1 x 1 x 1/2	1-1/2
1-1/4 x 1-1/4 x 1	1-7/8
1-1/4 x 1-1/4 x 3/4	1-7/8
1-1/4 x 1-1/4 x 1/2	1-7/8
1-1/2 x 1-1/2 x 1-1/4	2-1/4
1-1/2 x 1-1/2 x 1	2-1/4
1-1/2 x 1-1/2 x 3/4	2-1/4
1-1/2 x 1-1/2 x 1/2	2-1/4
2 x 2 x 1-1/2	2-3/8
2 x 2 x 1-1/4	2-1/4
2 x 2 x 1	2
2 x 2 x 3/4	1-3/4
2-1/2 x 2-1/2 x 2	2-3/4
2-1/2 x 2-1/2 x 1-1/2	2-5/8
2-1/2 x 2-1/2 x 1-1/4	2-1/2
2-1/2 x 2-1/2 x 1	2-1/4
3 x 3 x 2-1/2	3-1/4
3 x 3 x 2	3
3 x 3 x 1-1/2	2-7/8
3 x 3 x 1-1/4	2-3/4
3-1/2 x 3-1/2 x 3	3-5/8
3-1/2 x 3-1/2 x 2-1/2	3-1/2
3-1/2 x 3-1/2 x 2	3-1/4
3-1/2 x 3-1/2 x 1-1/2	3-1/8
4 x 4 x 3-1/2	4
4 x 4 x 3	3-7/8
4 x 4 x 2-1/2	3-3/4
4 x 4 x 2	3-1/2
4 x 4 x 1-1/2	3-3/8
5 x 5 x 4	4-5/8
5 x 5 x 3-1/2	4-1/2
5 x 5 x 3	4-3/8
5 x 5 x 2-1/2	4-1/4
5 x 5 x 2	4-1/8
6 x 6 x 5	5-3/8
6 x 6 x 4	5-1/8
6 x 6 x 3-1/2	5
6 x 6 x 3	4-7/8
6 x 6 x 2-1/2	4-3/4
8 x 8 x 6	6-5/8

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Table V - Reducing tee dimensions (all dimensions are in inches)(Cont'd)

IPS	M
8 x 8 x 5	6-3/8
8 x 8 x 4	6-1/8
8 x 8 x 3-1/2	6
10 x 10 x 8	8
10 x 10 x 6	7-5/8
10 x 10 x 5	7-1/2
10 x 10 x 4	7-1/4
12 x 12 x 10	9-1/2
12 x 12 x 8	9
12 x 12 x 6	8-5/8
12 x 12 x 5	8-1/2
14 x 14 x 12	10-5/8
14 x 14 x 10	10-1/8
14 x 14 x 8	9-3/4
14 x 14 x 6	9-3/8
16 x 16 x 14	12
16 x 16 x 12	11-5/8
16 x 16 x 10	11-1/8
16 x 16 x 8	10-3/4
16 x 16 x 6	10-3/8

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Table VI - Straight and reducing tee crotch thicknesses

Size Run x branch	Crotch minimum
2 x 2	0.190
2 x 1-1/2	.179
2 x 1-1/4	.169
2 x 1	.162
2 x 3/4	.157
2-1/2 x 2-1/2	.218
2-1/2 x 2	.204
2-1/2 x 1-1/2	.189
2-1/2 x 1-1/4	.185
2-1/2 x 1	.179
3 x 3	.280
3 x 2-1/2	.250
3 x 2	.239
3 x 1-1/2	.229
3 x 1-1/4	.224
3 x 1	.216
4 x 4	.356
4 x 3	.318
4 x 3-1/2	.333
4 x 2-1/2	.296
4 x 2	.285
4 x 1-1/2	.275
4 x 1-1/4	.269
5 x 5	.390
5 x 4	.373
5 x 3-1/2	.350
5 x 3	.335
5 x 2-1/2	.315
5 x 2	.304
6 x 6	.468
6 x 5	.429
6 x 4	.412
6 x 3-1/2	.390
6 x 3	.378
6 x 2-1/2	.363
6 x 2	.351
8 x 8	.630
8 x 6	.549
8 x 5	.523
8 x 4	.496
8 x 3-1/2	.483

All dimensions are in inches.

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Table VI - Straight and reducing tee crotch thicknesses (Cont'd)

Size Run x branch	Crotch minimum
8 x 3	.472
10 x 10	.710
10 x 8	.670
10 x 6	.589
10 x 5	.565
10 x 4	.540
12 x 12	.825
12 x 10	.805
12 x 8	.760
12 x 6	.680
12 x 5	.660

All dimensions are in inches

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Table VII - Elbow and return bend crotch thickness

Size	Thickness	
	Short radius	Long radius
1/2	0.085	0.080
3/4	.093	.087
1	.104	.096
1-1/4	.118	.108
1-1/2	.128	.117
2	.148	.133
2-1/2	.168	.151
3	.194	.173
3-1/2	.214	.190
4	.235	.208
5	.277	.245
6	.327	.282
8	.405	.352
10	.482	.426
12	.572	.496
14	.623	.540
16	.708	.610

Dimensions are in inches

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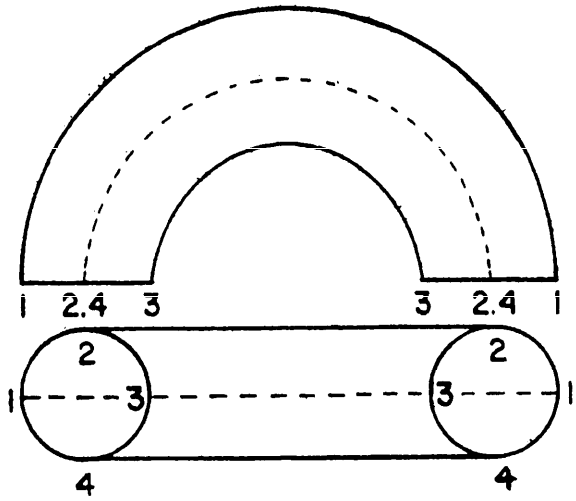
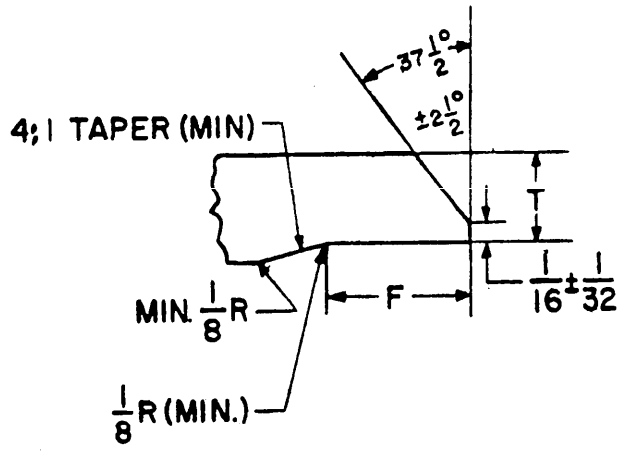


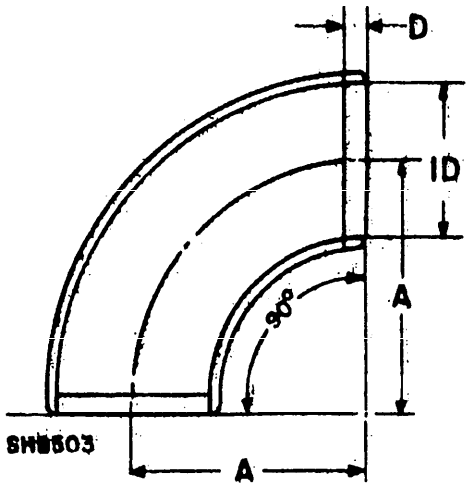
Figure 1 - Prohibited weld locations



NOTE: For other dimensions see table II.

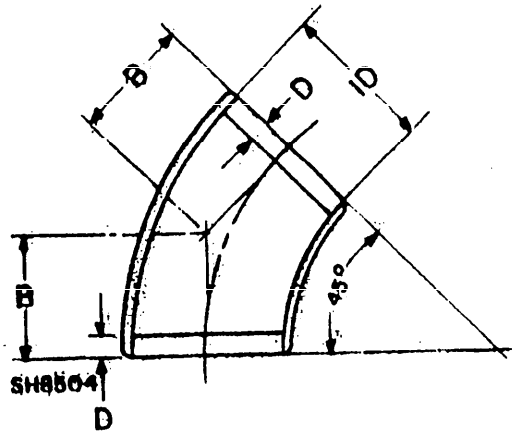
Figure 2 - Edge preparation for welding

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NOTE: For dimensions and tolerances, see tables II and III.

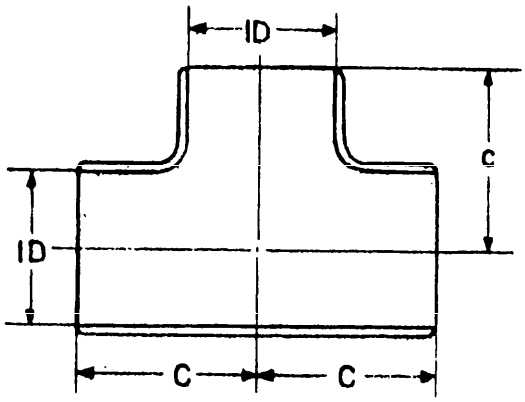
Figure 3 - Ninety-degree elbow



NOTE: For dimensions and tolerances, see tables II and III.

Figure 4 - Forty-five-degree elbow

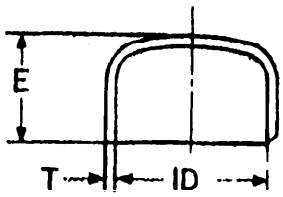
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SH8505

NOTE: For dimensions and tolerances, see tables II and III.

Figure 5 - Straight tee

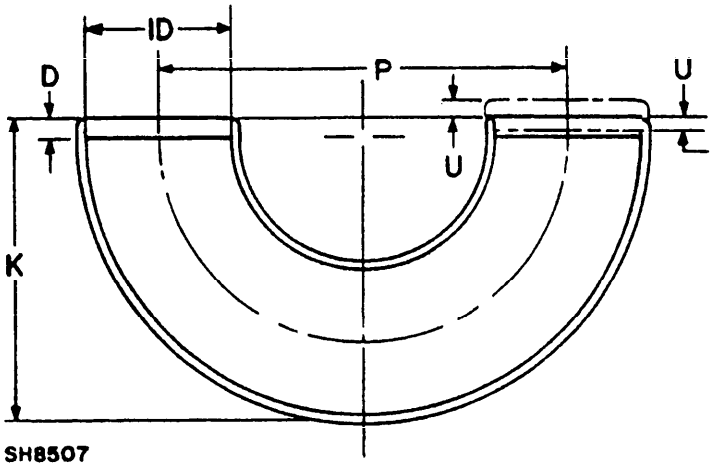


SH8506

NOTE: For dimensions and tolerances, see tables II and III.

Figure 6 - Cap

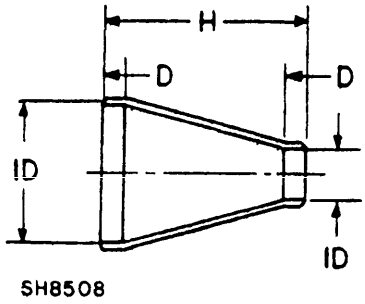
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NOTE: For dimensions and tolerances, see tables II and III.

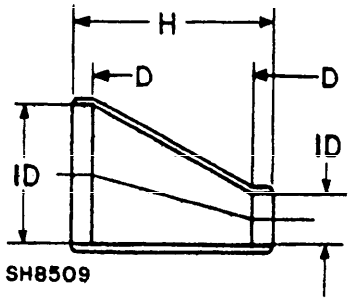
Figure 7 - One-Hundred-eighty-degree return bend

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NOTE: For dimensions and tolerances, see tables II, III and IV.

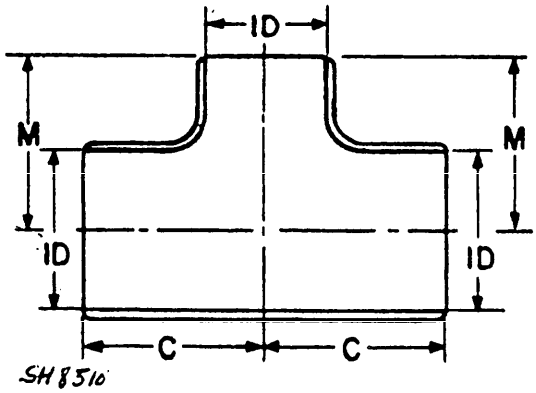
Figure 8 - Concentric reducers



NOTE: For dimensions and tolerances, see tables II, III and IV.

Figure 9 - Eccentric reducers

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NOTE: For dimensions and tolerances, see tables II, III and V.

Figure 10 - Reducing tees

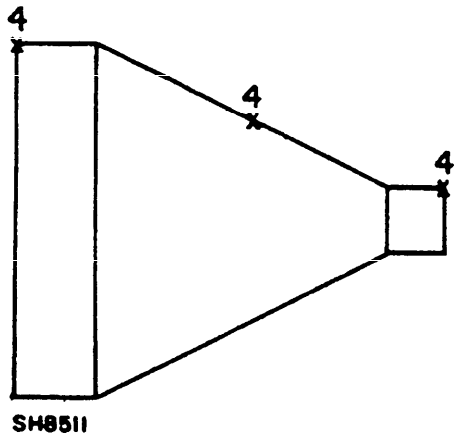


Figure 11 - Concentric reducer inspection points

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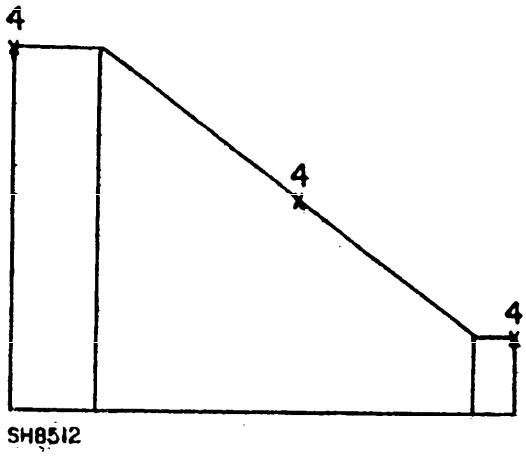


Figure 12 - Eccentric reducer inspection points

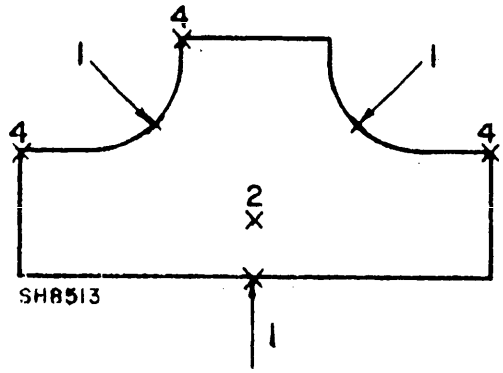


Figure 13 - Tee inspection points

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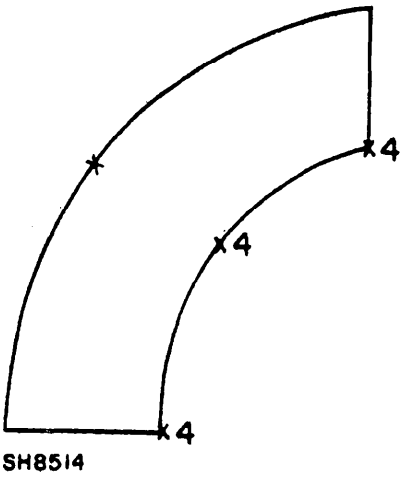


Figure 14 - Ninety-degree elbow inspection points

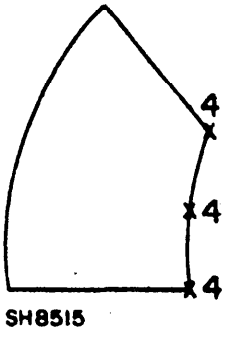


Figure 15 - Forty-five-degree elbow inspection points

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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